Chemistry Unit 1
Extended response #1

Bonding in Household Products

A wide range of chemicals and materials are used in and around the home. Materials used in the home have one of four types of bonding between particles: metallic, ionic, covalent molecular, or covalent network. Examples of some compounds is given below.

Metallic	Ionic	Covalent molecular	Covalent network
Steel Aluminium Copper	Sodium chloride Sodium hydrogen carbonate Calcium carbonate	Water Acetic acid Ammonia	Graphite Silicon dioxide Diamond

Your Task

- Research the uses of the compounds listed above. As you have studied the types of bonding and properties of substances in class, you do not need to research these but you will need to know about them to complete the in class component of the task.
- Prepare research notes about each compound. (See example)
- In class, in **Week 12** (= Week 2, Term 2), you will be asked to submit your notes and then complete the following task.

"The properties of many substances make them suitable to use around the home. Discuss the bonding, properties and common household uses of the following four compounds.

Compounds
To be advised

For each of the substances listed above describe:

- the type of bonding in each substance, including the arrangement of particles and the nature of the attractive forces that exist between particles
- the properties of the substance and an explanation of how these properties are related to the structure and bonding within the substance
- the uses of the substance in the home, and an explanation of how these uses are dependent on the properties of the substance.

- The four substances will be selected from the list above, but you will not know which will be selected prior to the essay.

Tips for Good Results

Notes	Essay
• Research the specific uses of each material	• Revise the structure/bonding of each type of substance
• Do not reference unreliable sites like Wikipedia, Wiki Answers or Yahoo Answers	• Plan your essay in advance
• Include formatted references for the sources of your information in your notes. See https://www.usq.edu.au/library/referencing/apa-referencing-guide for details.	• Aim to write at least half a page for each substance • Use headings and subheadings • Include diagrams when appropriate

Example Notes

Substance: Carbon dioxide

Uses

Fire suppression agent (e.g. fire extinguisher)

- Does not support combustion
- Is denser than air – will sink to ground
- Compressible (can be stored in a small volume container)

Soft drink additive

- Dissolves easily in water (compared to N_2 for example)
- Reacts with water to form carbonic acid (H2CO3) which improves taste of drink

Sources

Makowka, N. (n.d.) *Why carbon dioxide (CO_2) in fire suppression systems?* Retrieved March 3, 2014 from http://www.nafed.org/whyco2

Matson, B. (2006) *Microscale gas chemistry*. Retrieved from http://mattson.creighton.edu/Download_Folder/MattsonGasBook4thEdCO2.pdf

The Naked Scientists. (2008) Why do they use CO2 in fizzy drinks? Retrieved March 3, 2014 from http://www.thenakedscientists.com/HTML/questions/question/1952/

MARKING KEY

RESEARCH NOTES (to be handed in on the day of essay)

Description	Marks
Depth of research (e.g. detail)	2 marks
Quality of sources (reliability)	2 marks
Formatting of references uses a referencing system e.g. APA	2 marks

IN-CLASS ESSAY (no notes permitted during essay)

Description	Marks
Description of the type of bonding, including arrangement of particles and attraction between them	12 marks
Description of the properties of substances, and explanation of how those properties relate to the type of bonding	12 marks
Description of the uses of substances in the home, and explanation of how these uses is related to the properties of that compound	12 marks
Quality of essay (e.g. well organised, uses headings, minimal spelling errors)	3 marks